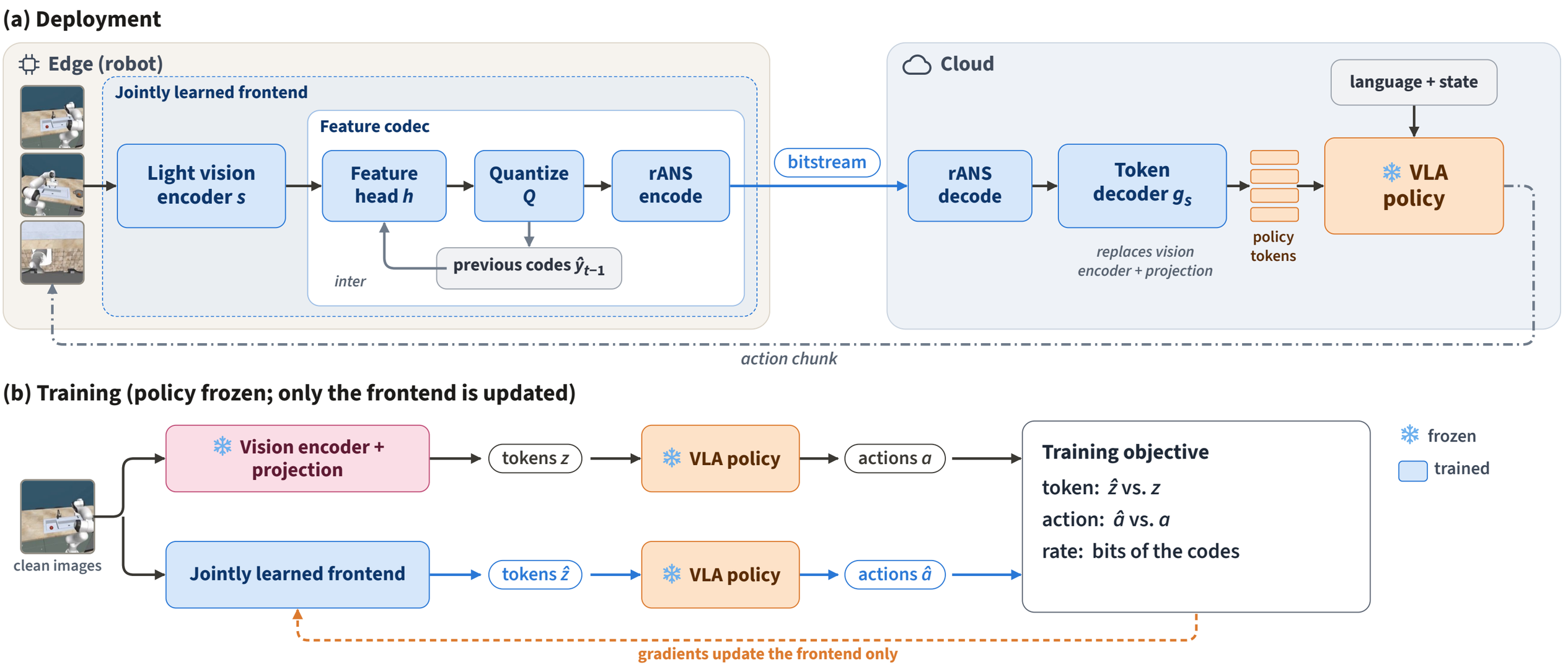

(a) Deployment
Edge (robot)
Jointly learned frontend
Feature codec
Light vision encoder s
Feature head h
Quantize Q
rANS encode
previous codes $\hat{y}_{t-1}$
inter
bitstream
Cloud
language + state
rANS decode
Token decoder g_s
replaces vision encoder + projection
policy tokens
VLA policy
action chunk
(b) Training (policy frozen; only the frontend is updated)
clean images
Vision encoder + projection
tokens z
VLA policy
actions a
Jointly learned frontend
tokens $\hat{z}$
VLA policy
actions $\hat{a}$
Training objective
token: $\hat{z}$ vs. z
action: $\hat{a}$ vs. a
rate: bits of the codes
frozen
trained
gradients update the frontend only